DERIVED PUSHFORWARD OF THE INTERNAL HOM AGAINST A PERFECT COMPLEX WITH PROPER COHERENT SUPPORT IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Duals of perfect complexes. Cohomology, lemma dual-perfect-complex (cross-chapter, named). E has a dual perfect complex E-dual.
- Tensor form of the pushforward perfectness. spaces-perfect.tex lemma-tensor-perfect (carded cross-chapter b45-era as S9-perf-tensor-perfect L26083; more general spaces version, same conclusion). Concludes the perfectness of K from the tensor identification.

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-perf2-ext-perfect/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0DJQ: sources/stacks-perfect/pinned-original-source.tex, source lines 7038–7038. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a Noetherian scheme. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $E \in D(\mathcal{O}_X)$ be perfect. Let $\mathcal{G}^\bullet$ be a bounded complex of coherent $\mathcal{O}_X$-modules flat over S with support proper over S. Then $K = Rf_*R\mathcal{H}om(E, \mathcal{G}^\bullet)$ is a perfect object of $D(\mathcal{O}_S)$.*

Proof. Since E is a perfect complex there exists a dual perfect complex $E^\vee$, see Cohomology, Lemma 08DQ. Observe that $R\mathcal{H}om(E, \mathcal{G}^\bullet) = E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet$. Thus the perfectness of K follows from Lemma 0DJQ. □

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.